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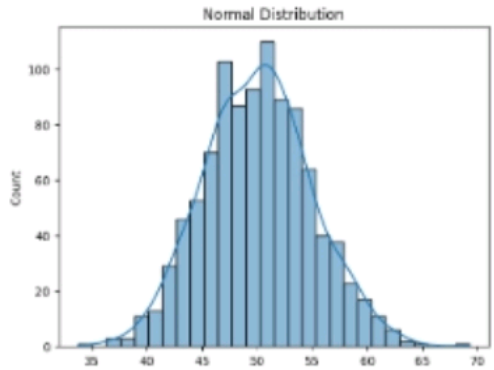
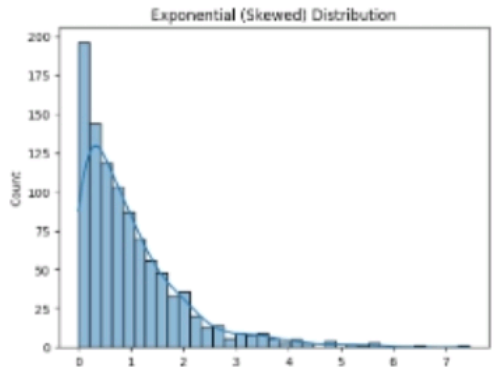
Welcome to Colab!

```python
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import skew, kurtosis, zscore

# Generate Data
np.random.seed(42)

normal_data = np.random.normal(loc=50, scale=5, size=1000)
skewed_data = np.random.exponential(scale=1, size=1000)

# Visualize
plt.figure()
sns.histplot(normal_data, kde=True)
plt.title("Normal Distribution")
plt.show()

plt.figure()
sns.histplot(skewed_data, kde=True)
plt.title("Exponential (Skewed) Distribution")
plt.show()

# Calculate Skewness
print("Skewness (Normal Data):", skew(normal_data))
print("Skewness (Skewed Data):", skew(skewed_data))

# Calculate Kurtosis (Fisher's definition - excess kurtosis)
print("Kurtosis (Normal Data):", kurtosis(normal_data))
print("Kurtosis (Skewed Data):", kurtosis(skewed_data))

# Outlier Detection using Z-score
def detect_outliers(data):
    z_scores = zscore(data)
    outliers = data[np.abs(z_scores) > 3]
    return outliers

outliers = detect_outliers(skewed_data)
print("Number of Outliers in Skewed Data:", len(outliers))
```






```
Skewness (Normal Data): 0.11688083110533704
Skewness (Skewed Data): 1.9808119458203124
Kurtosis (Normal Data): 0.06620589292148482
```

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```python
import numpy as np
import pandas as pd
from scipy.stats import zscore

# Given house prices (in $100,000s)
prices = np.array([2.5, 3.0, 2.8, 3.2, 15.0, 2.9, 3.1, 2.7, 100.0])

# Convert to DataFrame
df = pd.DataFrame({"House_Price": prices})

print("Original Data:\n", df)

# Step 1: Detect Outliers using Z-score
z_scores = zscore(df["House_Price"])
df["Z_score"] = z_scores

outliers = df[np.abs(df["Z_score"]) > 3]

print("\nDetected Outliers:\n", outliers)

# Step 2: Treat the data entry error (100.0 should be 10.0)
df.loc[df["House_Price"] == 100.0, "House_Price"] = 10.0

print("\nData After Correcting 100.0 to 10.0:\n", df)

# Step 3: Recalculate Z-score after correction
df["Z_score"] = zscore(df["House_Price"])

print("\nUpdated Data with New Z-scores:\n", df)

# Step 4: Summary Statistics
print("\nMean Price:", df["House_Price"].mean())
print("Median Price:", df["House_Price"].median())
```

```
Original Data:
    House_Price
0          2.5
1          3.0
2          2.8
3          3.2
4         15.0
5          2.9
6          3.1
7          2.7
8        100.0

Detected Outliers:
 Empty DataFrame
Columns: [House_Price, Z_score]
Index: []

Data After Correcting 100.0 to 10.0:
    House_Price   Z_score
0          2.5 -0.413529
1          3.0 -0.397018
2          2.8 -0.403622
3          3.2 -0.390413
4         15.0 -0.000734
5          2.9 -0.406320
6          3.1 -0.393715
7          2.7 -0.406925
8         10.0  2.806275

Updated Data with New Z-scores:
    House_Price   Z_score
0          2.5 -0.604588
1          3.0 -0.584736
2          3.8 -0.532677
3          3.2 -0.436795
4         15.0  2.391718
5          2.9 -0.588786
6          3.1 -0.468765
7          2.7 -0.556647
8         10.0  1.193195

Mean Price: 5.022222222222222
Median Price: 3.0
```